Aerial block for Thursday 20 August 2026: Westbrook

To Kristine Kirby, Operations Manager. From Kyle Blair, Surveillance Coordinator. Monday 17 August 2026, on the surveillance of MMWR week 32 (trap-nights of 10 and 11 August).

Fly Westbrook. It had the most West Nile virus-infected Culex females per CO2 trap-night last week: an estimated 2.3. The runner-up is South Prairie at 1.8. Westbrook keeps the flight unless the infection of its CO2-caught Culex females is below 18 per 1,000; last week it was 23.5 per 1,000.

How the figure is built

For each zone, the Culex tarsalis and Culex pipiens females caught per CO2 trap-night that ran (failed traps left out), times the share of those same females infected, summed over the two species. The infected share comes from the pools made from the CO2 traps' own catch, estimated by maximum likelihood for pools of varying size. The bias-corrected estimate gives the same figures to one decimal.

Two shortcuts give the wrong zone. Counting each positive pool as one infected mosquito (the minimum infection rate) understates our laboratory's pools of 50: 42 of Westbrook's 70 Culex tarsalis pools were positive, and a pool that full of positives holds more than one infected female. And the gravid traps in Old Town and Riverside South catch females that have already fed; their pools run several times hotter than the host-seeking females the CO2 traps catch, so they cannot stand in for them. Coulee Hills has the most human cases this season, but they fell ill between late June and late July, in the coulee focus that was flown on 16 and 23 July.

The two leading zones, week 32

	Westbrook	South Prairie
CO2 trap-nights run	30	24
Culex tarsalis females per trap-night	95.2	134.9
Culex pipiens females per trap-night	3.8	8.2
Culex tarsalis pools (positive)	70 (42)	334 (41)
Culex pipiens pools (positive)	25 (2)	33 (0)
Culex tarsalis infected per 1,000	23.7	13.4
Culex pipiens infected per 1,000	18.2	0.0
Infected Culex females per CO2 trap-night	**2.3**	**1.8**

Source: field database extract of 17 August 2026, 08:20 (CO2 trap-nights, catches and the district laboratory's results) and the county laboratory's report of 14 August 2026, whose week-32 results are not yet keyed into the database. Infection estimated by maximum likelihood from each zone's CO2-trap pools, by species.

The flip point is South Prairie's figure over Westbrook's 99.0 Culex females per CO2 trap-night. The season's human cases and trap-nights by zone are in zone_season_2026.xlsx, and the cases by week of onset in human_cases_by_zone.png.

Neither zone lost a trap-night to a failure in week 32 (protocol 3.4), and neither had an inconclusive pool. Designation to Ostby Aviation by 17:00 today (agreement clause 2.2).